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Exp_8

Title: Dijkstra's Algorithm

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In [3]: n = int(input("Enter number of vertices in the graph: "))

print("Enter the adjacency matrix where each row is separated by spaces:")
graph = []
for i in range(n):
    row = input().split()
    graph.append([int(x) for x in row])

source = int(input("Enter the source vertex: "))

def dijkstra(graph, source):
    V = len(graph)

    dist = [float('inf')] * V
    dist[source] = 0

    visited = [False] * V

    for _ in range(V):
        # Find the vertex with the smallest distance that has not been visited
        min_dist = float('inf')
        min_index = -1

        for v in range(V):
            if not visited[v] and dist[v] < min_dist:
                min_dist = dist[v]
                min_index = v

        # Mark the selected vertex as visited
        visited[min_index] = True

        # Update distance values of the adjacent vertices
        for v in range(V):
            if not visited[v] and graph[min_index][v]:
                dist[v] = min(dist[v], dist[min_index] + graph[min_index][v])

    return dist

Shortest_Paths = dijkstra(graph, source)
print("=====")
print("Shortest Distance from node", source, "to each node: ")
for i, j in enumerate(Shortest_Paths):
    print("Vertex", i, "distance = ", j)
print("=====")
```

Enter the adjacency matrix where each row is separated by spaces:

```
=====
Shortest Distance from node 0 to each node:
Vertex 0 distance = 0
Vertex 1 distance = 9
Vertex 2 distance = 13
Vertex 3 distance = 7
Vertex 4 distance = 19
=====
```